

Project Synopsis

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Course: B.Tech (Computer Science Engineering) Project

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Project Title: Bizlens: SMB's Decision Support System

Abstract

Small and Medium-sized Businesses (SMBs) generate and maintain a considerable amount of operational information through products, inventory, customers, orders and business expenses. However, such information is often maintained through separate records, spreadsheets or disconnected systems, making it difficult for business owners to obtain a consolidated view of business performance and make informed decisions.

BizLens is a full-stack web-based business management and analytics platform designed to integrate these operational activities into a centralized system. The platform enables SMB users to manage products, inventory, customers, orders and expenses while transforming the recorded data into meaningful business information through KPI-based analytics and decision-support features.

The system provides analytical measures such as revenue, product cost, gross profit, operating expenses, net profit, net profit margin, average order value, order performance, customer activity and product performance. It also supports period-based comparisons, expense analysis, simplified break-even analysis and rule-based business alerts to help users identify important changes and areas requiring attention.

The application is developed using **Node.js** and **Express.js** for the backend, **MySQL** for relational data management, and **HTML, CSS and JavaScript** for the browser-based frontend. Additional capabilities include secure authentication, inventory consistency through order transactions, order cancellation with stock restoration, business data export, responsive access, theme support and multilingual interface capabilities.

The primary objective of BizLens is to demonstrate how a full-stack information system can combine operational business data, business rules, KPI-based analytics and decision-support mechanisms within a single platform for SMBs.

Literature Review

Business Intelligence and Analytics (BI&A) has become an important area of research as organizations increasingly rely on data to understand business performance and support decision-making.

Chen, Chiang and Storey presented Business Intelligence and Analytics as a broad research area covering its evolution, applications and analytical capabilities. Their work establishes the importance of transforming large volumes of business data into meaningful information for organizational decision-making.

Watson and Wixom discussed Business Intelligence as a process involving the collection, integration and delivery of information for reporting, analysis and decision-making. This concept is reflected in BizLens through the integration of products, customers, orders and expenses into a centralized business information system.

Popović et al. examined the relationship between BI system maturity, information quality, analytical decision-making culture and the effective use of information. Their research highlights the importance of structured and reliable information for analytical decision-making. BizLens addresses this principle by maintaining structured operational data and applying defined business rules and KPI calculations.

Işık, Jones and Sidorova investigated BI capabilities and decision environments and identified factors such as data quality, integration and accessibility as important elements of BI success. BizLens follows a similar approach by integrating multiple business modules and providing users with direct access to analytical information.

Elbashir, Collier and Davern studied the relationship between Business Intelligence systems, business-process performance and organizational performance. Their findings emphasize the importance of connecting business processes with measurable performance indicators. BizLens applies this concept by connecting operational records with measures such as revenue, product cost, expenses, profit, margins and customer activity.

Fink, Yogev and Even examined how Business Intelligence resources and capabilities contribute to value creation and organizational learning. Their

work demonstrates that BI systems can provide value beyond basic reporting by supporting interpretation and organizational decision-making. BizLens similarly attempts to transform basic business records into actionable business information.

Llave conducted a systematic literature review on Business Intelligence and Analytics in Small and Medium-sized Enterprises. The study discusses BI&A solutions, adoption, implementation and benefits within the SME context, making it particularly relevant to the target environment of BizLens.

Wee, Scheepers and Tian investigated how Small and Medium Enterprises derive business value from Business Intelligence and Analytics. Their work supports the idea that appropriate analytical systems can help smaller organizations obtain value from their operational data.

Overall, the reviewed literature indicates that effective Business Intelligence involves more than storing operational data. Data integration, information quality, accessibility, analytical capabilities and decision support are important for converting business information into useful business insights. BizLens applies these principles within an academic full-stack implementation targeted at SMBs.

Problem Formulation & Planning

3.1 Problem Statement

Small and Medium-sized Businesses generate operational information through sales, products, inventory, customers, orders and expenses. However, these records may often be maintained through separate files, spreadsheets, manual records or disconnected systems.

As a result, business owners may face difficulties in obtaining a consolidated understanding of:

- Revenue
- Product costs
- Expenses
- Profitability
- Product performance
- Inventory status
- Customer activity
- Period-wise business performance

Manual analysis of these records can also require additional time and effort. At the same time, many enterprise-level business management and analytics solutions may provide functionality beyond the requirements of smaller businesses, making them comparatively complex for smaller organizations. Therefore, the problem addressed by BizLens is the development of an integrated web-based system that can **manage core SMB operational data and transform it into understandable KPIs, analytics, business insights and decision-support information.**

3.2 Objectives

The major objectives of BizLens are:

1. To develop a full-stack web application for managing products, inventory, customers, orders and business expenses.
2. To provide a centralized MySQL-based relational database for storing and managing business information.
3. To implement secure user authentication and business-profile management.
4. To provide KPI-based business analytics covering revenue, product cost,

gross profit, expenses, net profit, profit margin, orders and customer activity.

5. To provide product and customer performance analysis based on recorded business data.
6. To provide period-based analytics and comparisons to identify changes in business performance.
7. To implement defined business rules for profit, margin, inventory and business alerts.
8. To provide simplified break-even analysis based on available business data.
9. To generate rule-based business insights from operational and analytical data.
10. To maintain inventory consistency during order creation, modification and cancellation.
11. To provide business data export functionality in supported formats.
12. To evaluate the developed system through functional testing and realistic business scenarios.

3.3 Scope

The scope of BizLens includes the development of a full-stack web-based business information and analytics platform for Small and Medium-sized Businesses.

A. Authentication and Account Management

- User registration
- User login
- Password management
- Secure authentication
- Business profile management
- JWT-based authentication

B. Product and Inventory Management

- Product creation
- Product editing
- Product deletion/inactivation
- Product categories
- Cost price
- Selling price

- Stock quantity
- Product status
- Stock updates through order transactions
- Low-stock identification

C. Customer Management

- Customer information
- Customer history
- Order history
- New customer classification
- Returning customer classification
- Customer activity analysis

D. Order Management

- Creation of multi-product orders
- Order details
- Quantity management
- Order editing
- Order cancellation
- Order status
- Order history
- Automatic stock adjustment
- Stock restoration after cancellation

E. Expense Management

- Expense recording
- Expense categories
- Expense editing
- Expense deletion
- Expense date management
- Expense analysis

F. Business Analytics

- Revenue analysis
- Product cost analysis
- Gross profit
- Operating expense analysis
- Net profit
- Net profit margin

- Average Order Value
- Order analysis
- Customer activity analysis
- Product performance analysis
- Expense analysis
- Period comparison
- Expense-to-revenue ratio
- Simplified break-even analysis

G. Business Insights and Alerts

- Rule-based business insights
- Low-margin alerts
- Low-stock alerts
- Loss-making product warnings
- Expense-related alerts
- Performance comparison insights

H. Data Export

- Business data export
- CSV export
- Supported spreadsheet-format export

I. User Interface

- Responsive interface
- Existing BizLens visual design
- Light and dark themes
- Multilingual interface support

3.4 Business Rules and KPI Framework

One of the important components of BizLens is the transformation of operational records into measurable business indicators.

Revenue

$$\text{Revenue} = \Sigma(\text{Selling Price} \times \text{Quantity})$$

Product Cost

$$\text{Product Cost} = \Sigma(\text{Cost Price} \times \text{Quantity})$$

Gross Profit

Gross Profit = Revenue – Product Cost

Net Profit

Net Profit = Gross Profit – Operating Expenses

Net Profit Margin

Net Profit Margin = (Net Profit / Revenue) × 100

Average Order Value

AOV = Total Revenue / Number of Valid Orders

Expense-to-Revenue Ratio

Expense Ratio = (Total Expenses / Revenue) × 100

Period Growth

Growth % = ((Current Period – Previous Period) / Previous Period) × 100

The system should handle zero denominators safely and should not produce invalid values such as NaN or Infinity.

These KPIs form the foundation for the dashboard, analytics and business insight generation.

3.5 Out of Scope

BizLens is not intended to function as a complete enterprise accounting or financial-management system.

The following are outside the scope:

- Statutory accounting
- Tax filing
- Payroll processing
- Banking integration
- Payment-gateway settlement
- Audited financial statements
- Professional financial advice

- Advanced enterprise resource planning
- Complex warehouse management

This scope limitation keeps the system focused on **business data management, analytics and decision support for SMBs.**

Project Planning & Methodology

The project follows an **incremental software development methodology**, where requirements, system design, implementation, integration and testing are completed through successive phases.

Phase 1 — Requirement Analysis and Planning

- Problem identification
- Requirement gathering
- Stakeholder identification
- Literature review
- Feature identification
- Business-rule identification
- KPI definition
- Initial project planning

Phase 2 — System Design

- System architecture
- Database design
- ER model
- API planning
- User interface structure
- Workflow design
- Business logic design

Phase 3 — Implementation

- Frontend development
- Backend development
- Authentication
- MySQL integration
- Product management
- Customer management
- Order management
- Expense management
- Analytics implementation
- Business insight implementation

Phase 4 — Integration and Testing

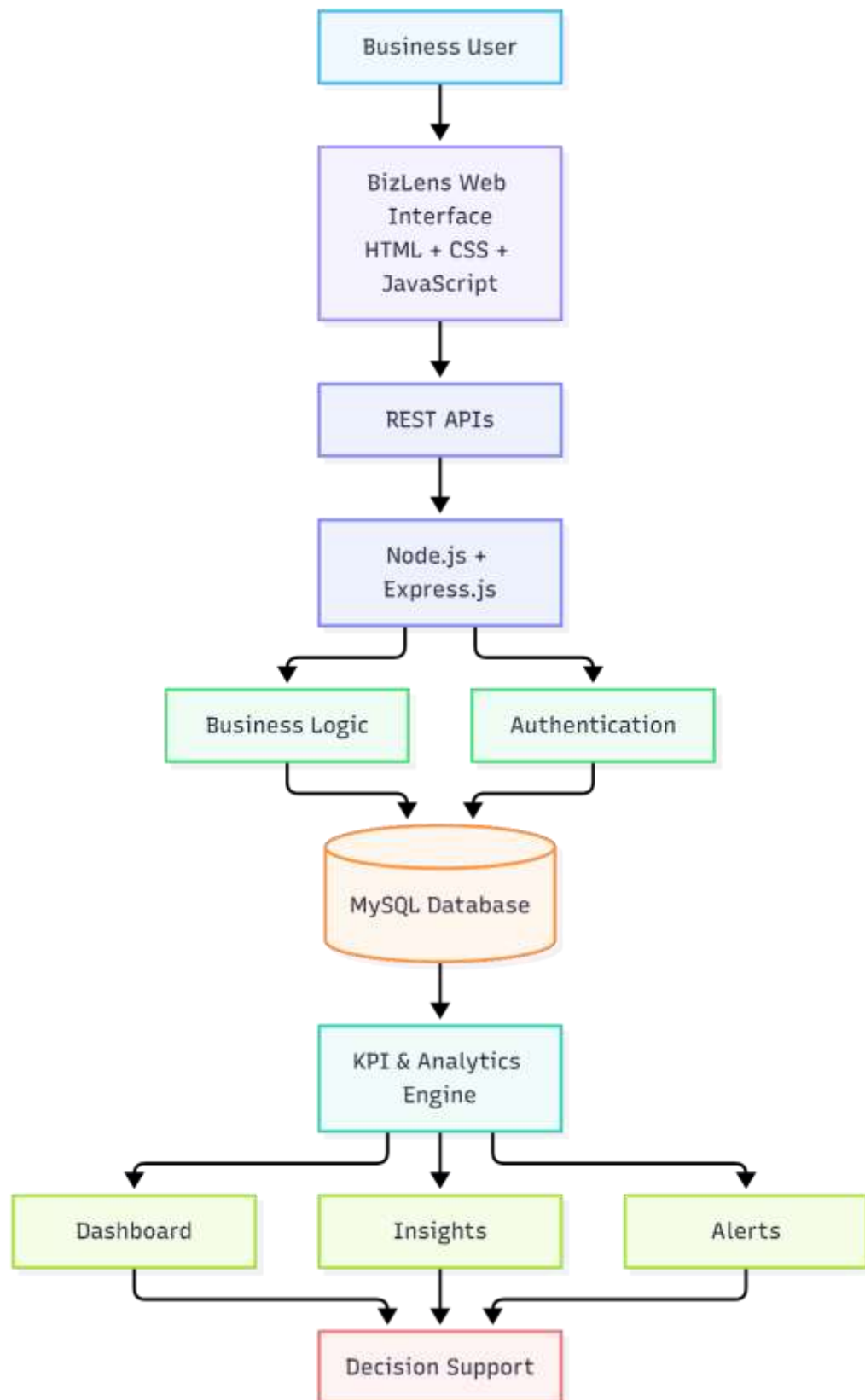
- Module integration
- Functional testing
- KPI validation
- Business-rule testing
- Inventory consistency testing
- Authentication testing
- Database testing
- Error-handling validation

Phase 5 — Finalization

- UI refinement
- Final testing
- Documentation
- Project demonstration
- Deployment preparation
- Final synopsis/report preparation

SYSTEM ARCHITECTURE

The proposed architecture follows a layered full-stack approach:



This architecture separates presentation, application logic, database management and analytics.

Expected Deliverables

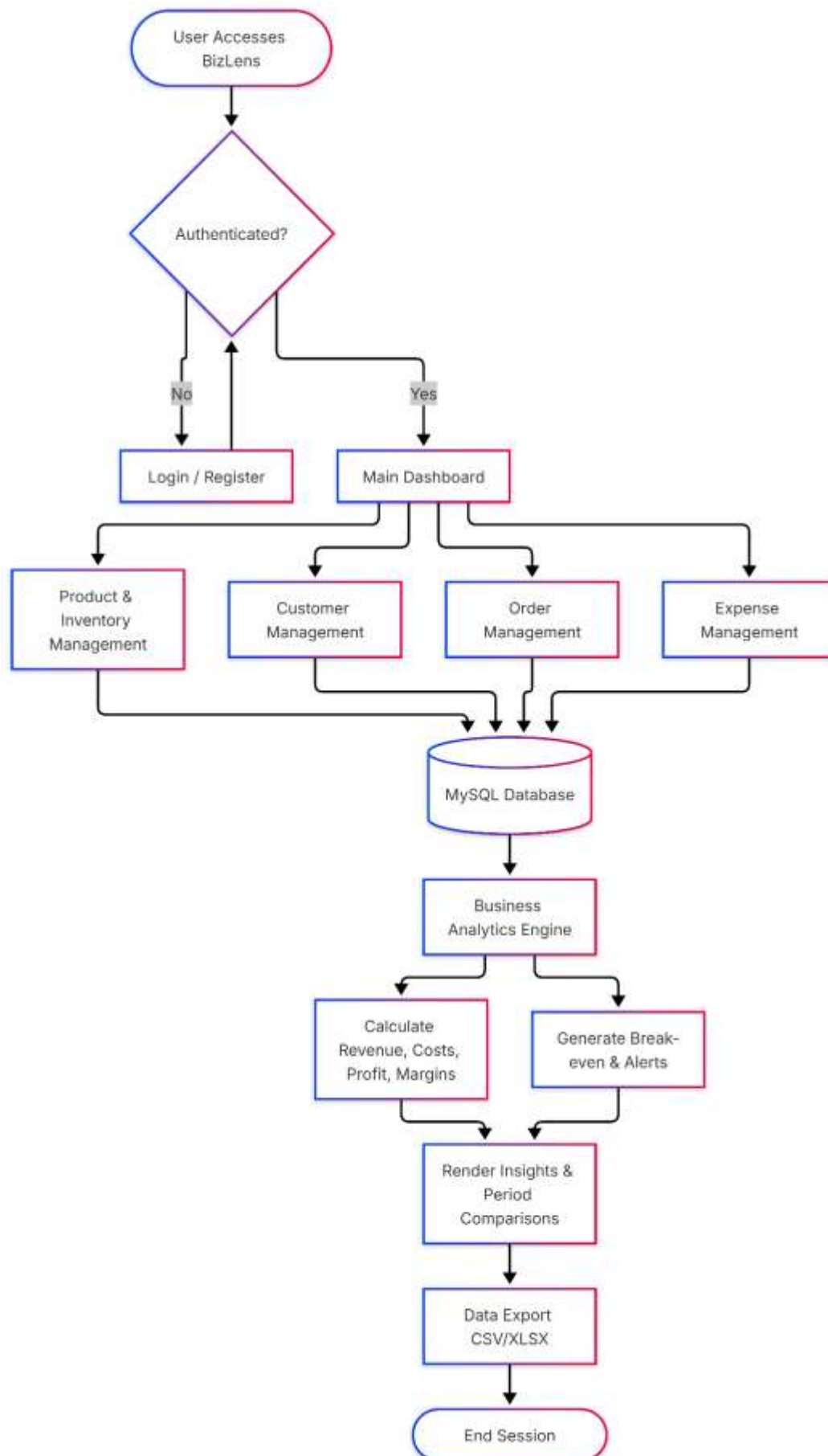
The expected deliverables of the project are:

1. A functional full-stack BizLens web application.
2. MySQL database schema.
3. Responsive frontend interface.
4. Authentication and account-management module.
5. Product and inventory management module.
6. Customer management module.
7. Order management module.
8. Expense management module.
9. Dashboard and KPI analytics module.
10. Business Insights module.
11. Business alert functionality.
12. Simplified break-even analysis.
13. REST API services.
14. Business rules and KPI definitions.
15. Data export functionality.
16. Functional testing and validation.
17. System architecture and database documentation.
18. Final project documentation and demonstration.

Anticipated Challenges & Proposed Solutions

Challenge	Proposed Solution
Maintaining consistency between orders and inventory	Use database transactions, stock validation and stock restoration during cancellation
Correctness of business calculations	Define KPI formulas clearly and validate calculations using controlled test data
Authentication and sensitive configuration	Use password hashing, JWT-based authentication and environment variables
Integration of multiple modules	Use structured REST APIs and consistent frontend-backend communication
Analytics for different reporting periods	Implement explicit date-range handling and period comparison logic
Usability for non-technical users	Maintain a simple dashboard, clear labels, responsive layouts and intuitive navigation
Maintaining historical order accuracy	Store transaction-level price and cost information with order items
Testing realistic business scenarios	Use demo data and scenario-based testing covering completed, pending and cancelled orders

System Flowchart



Conclusion

BizLens proposes a practical full-stack business information and analytics platform for Small and Medium-sized Businesses. The system integrates operational business activities such as product management, inventory, customers, orders and expenses within a centralized platform.

Beyond basic data management, BizLens applies defined business rules and KPI calculations to transform operational records into analytical information. Through dashboards, period comparisons, product and customer analysis, expense analysis, business alerts and simplified break-even analysis, the system aims to provide SMB users with a clearer understanding of their business performance.

The project therefore demonstrates the integration of **full-stack web development, relational database management, business analytics and decision-support concepts** within a single academic software system.

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